

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	09 july 2025
Team ID	
Project Name	DarshanEase
Maximum Marks	

Technical Architecture:

DarshanEase follows a modern three-tier web architecture consisting of the presentation layer, application layer, and data layer.

1. **Presentation Layer (Frontend)** The user interacts with the system through a responsive web interface built using React.js and Tailwind CSS. This layer allows devotees to explore temples, book Darshan slots, and make donations easily.
2. **Application Layer (Backend)** The backend is developed using Node.js and Express.js, which manages business logic such as user authentication, temple management, slot booking, and payment processing.
3. **Data Layer (Database)** The system stores application data in MongoDB, which manages user accounts, temple information, booking records, and donation data.
4. **External Services** The system integrates with payment gateway APIs to handle secure online donations and VIP Darshan payments.
5. **Infrastructure Layer** The platform can be deployed on cloud infrastructure or local servers, ensuring scalability and availability for large numbers of devotees.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interface through which devotees and organizers interact with the system	React.js, Tailwind CSS, HTML, JavaScript
2.	Application Logic-1	User authentication and authorization logic	Node.js, Express.js, JWT
3.	Application Logic-2	Darshan slot booking and temple management logic	Node.js, Express.js
4.	Application Logic-3	Payment processing and donation management	Payment Gateway API
5.	Database	Stores user data, temple details, bookings, and donations	MongoDB
6.	Cloud Database	Cloud database hosting service	MongoDB Atlas
7.	File Storage	Storage for temple images and media files	Cloud Storage / Local File System
8.	External API-1	Payment gateway integration for online donations	Razorpay / Stripe API
9.	External API-2	Identity verification or government APIs if required	Aadhaar API (optional)
10.	Machine Learning Model	Purpose of Machine Learning Model	Object Recognition Model, etc.
11.	Infrastructure (Server / Cloud)	Application deployment environment	Node Server, Cloud Hosting (AWS / Vercel / Render)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used to build the application	React.js, Node.js, Express.js
2.	Security Implementations	Secure authentication and data protection mechanisms	JWT Authentication, Bcrypt Encryption, HTTPS
3.	Scalable Architecture	System designed to support growing users and temples	MERN Stack Architecture

S.No	Characteristics	Description	Technology
4.	Availability	System available continuously for devotees	Cloud Deployment, Load Balancing
5.	Performance	Fast response and efficient booking system	Optimized APIs, Database Indexing, CDN

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-temple-ticket-booking/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>